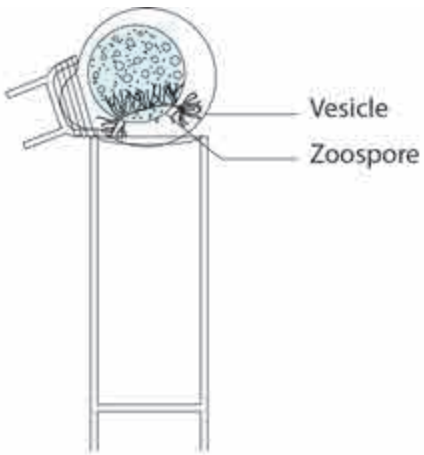
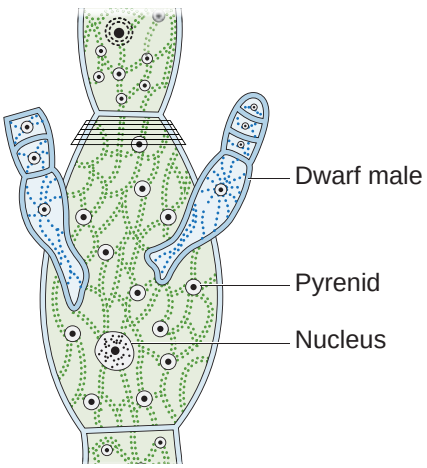
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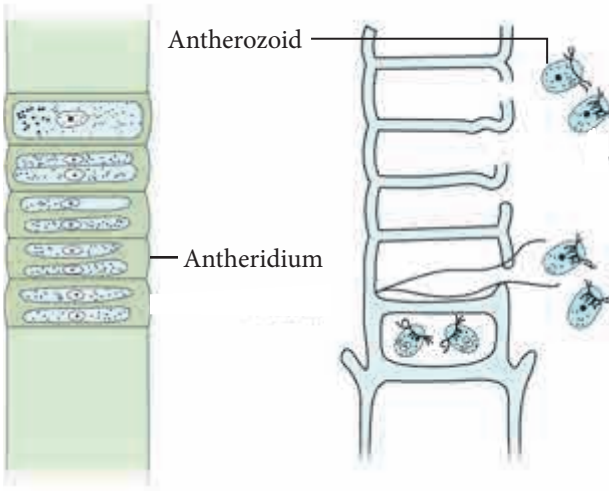
Oedogonium reproduces by vegetative, asexual and sexual methods. Vegetative reproduction takes place by fragmentation and akinete formation. During asexual reproduction zoospores are formed. During favourable conditions, some of the vegetative cells function as zoosporangia. Usually a single zoospore is produced per zoosporangium. A ring of short flagella is found at the base of colourless, beak like anterior end of the zoospore. This kind of flagellation is called stephanokont. The zoospore is released from the zoosporangium and swims in water (Figure 2.6). If it reaches a suitable substratum, it divides into two cells. The lower cell forms holdfast. The green upper cell divides and produces the filament.



a) Zoospore formation



b) Dwarf male



c) Filament showing antheridium

Figure 2.6: Reproduction in *Oedogonium*

Sexual reproduction is Oogamous. The male gametangium is antheridium and female gametangium is called Oogonium. Based on the distribution of sex organs there are two types of species namely Macrandrous and Nannandrous.

Macrandrous monoecious – Antheridia and Oogonia occur on same filament – *Oedogonium fragile*.

Macrandrous dioecious – Antheridia and Oogonia occur on separate filaments – *Oedogonium crassum*

In nannadrous species antheridia are produced on reduced male filaments called dwarf male plants(*O.cancatenatum*).

In nannandrous species antheridia develop on specialised 2–4 celled filaments called dwarf males. The dwarf male is developed from androspores released from the androsporangium.. If the androsporangia and oogonia develop on same filament, it is called **gynandrosporous** (*O. concatenatum*). If they are borne on different filaments it is called **idioandrosporous** (*O. conferatum*). The antheridium produces multiflagellate antherozoids. They are released by